

LATTICE ICE™ Technology Library

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Revision History

Version	Changes
2.0	Added Version Number to document. Added sections on Default Signal Values for unconnected ports.
2.1	Added PLL primitives
2.2	Corrected SB_CARRY connections to LUT inputs
2.3	Added iCE40 RAM, PLL primitives.
2.4	Added PLL_DS, SB_MIPi_RX_2LANE, SB_TMDS_deserializer primitives.
2.5	Added SB_MAC16 Primitive details.
2.6	Added iCE40LM Hard Macro details. Removed PLL_DS, SB_MIPi, SB_TMDS, SB_MAC16 primitive details.
2.7	Added iCE5LP (iCE40 Ultra) primitive details.
2.8	Removed iCE65 RAM, PLL details.
2.9	Added iCE40UL (iCE40 Ultra Lite) primitive details
3.0	Added iCE40UP (iCE40 Ultra Plus) primitives.

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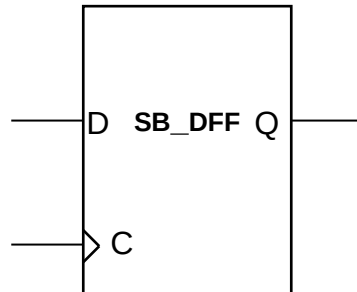
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Register Primitives

SB_DFF

D Flip-Flop

Data: D is loaded into the flip-flop during a rising clock edge transition.



Inputs			Output
	D	C	Q
	0	↗	0
	1	↗	1
Power on State	X	X	0

Key

↗ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL use

This register is inferred during synthesis and can also be explicitly instantiated.

Verilog Instantiation

```
// SB_DFF - D Flip-Flop.
SB_DFF SB_DFF_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .D(D),           // Data
);

// End of SB_DFF instantiation
```

VHDL Instantiation

```
-- SB_DFF - D Flip-Flop.
```

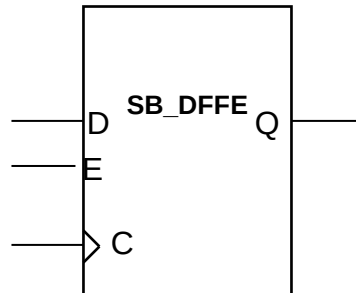
```
SB_DFF_inst: SB_DFF
  port map (
    Q => Q,          -- Registered Output
    C => C,          -- Clock
    D => D,          -- Data
  );

-- End of SB_DFF instantiation
```


SB_DFFE

D Flip-Flop with Clock Enable

Data D is loaded into the flip-flop when Clock Enable E is high, during a rising clock edge transition.



Inputs			Output
E	D	C	Q
0	X	X	Previous Q
1	0	↗	0
1	1	↗	1
Power on State	X	X	0

Key

- ↗ Rising Edge
- 1 High logic level
- 0 Low logic level
- X Don't care
- ? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'

Input C: Logic '0'

Input E: Logic '1'

Note that explicitly connecting a logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the logic '1'. It is recommended that the user leave the port E unconnected, or use the corresponding flip-flop without Enable functionality i.e. the DFF primitive.

Verilog Instantiation

```
// SB_DFFE - D Flip-Flop with Clock Enable.
```

```
SB_DFFE      SB_DFFE_inst (
    .Q(Q),           // Registered Output
    .C(C),           // Clock
    .D(D),           // Data
    .E(E),           // Clock Enable
);
```

```
// End of SB_DFFE instantiation
```

VHDL Instantiation

-- SB_DFFE - D Flip-Flop with Clock Enable.

```
SB_DFFE_inst: SB_DFFE
    port map (
        Q => Q,          -- Registered Output
        C => C,          -- Clock
        D => D,          -- Data
        E => E,          -- Clock Enable
    );
```

-- End of SB_DFFE instantiation